

$$|SL(2, p)| = |GL(2, p)| / (p-1) = \dots$$

Homework 4. Group Theory Stone Chan.

Problem 1.

a) in 4 entries we can fill with $\{0, 1, 2, \dots, p-1\}$ number, but the determinant has to be nonzero, so (a, b) shall be linearly independent of (c, d) , where we demand one of c, d shall be nonzero as well

$$|GL(2, p)| = \frac{(p^2-1)(p^2-p)}{p-1} = (p^2-1)(p-1)$$

$\uparrow$ $\uparrow$
 c d has to be nonzero, can't be simultaneously 0

$\rightarrow$ p situations where $(ab) = K(c, d)$
 $K = 0, 1, 2, \dots, (p-1)$

b) to show it forms a subgroup, we need to prove closure

pick $\forall A, B \in SL(2, p)$ $C = AB, C \in GL(2, p)$

and $\det(C) = \det(A) \det(B) = 1$

so $SL(2, p)$ forms a subgroup.

$\det m$ can take value $1, 2, 3, \dots, p-1$. now since $SL(2, p)$

is a subgroup, also for $\forall g \in GL(2, p), \forall h \in SL(2, p)$

~~$g \in SL(2, p)$~~ $ghg^{-1} \in SL(2, p)$ because $\det(ghg^{-1}) = 1$

$SL(2, p)$ is a normal Subgroup.

then we can classify G by $SL(2, p)$

$G/SL(2, p) = \{M_1, M_2, \dots, M_{p-1}\}$ where M_k ~~represents~~ is matrix determinant k representative.

e) $|GL(2,2)| = 3 \times 2 = 6$

GL2 $|SL(2,p)| = |GL(2,p)| / (p-1) = p(p^2-1)$

c). the center of $GL(2,p)$ shall be

$$C(GL(2,p)) = \left\{ \begin{pmatrix} \lambda & 0 \\ 0 & \lambda \end{pmatrix} : \lambda \in \mathbb{Z}_p^* \right\}$$

• $C(SL(2,p))$ shall be matrices in $C(GL(2,p))$ also with determinant 1, $\left\{ \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}, \begin{pmatrix} p-1 & 0 \\ 0 & p-1 \end{pmatrix} \right\}$

d) $PSL(2,p) = SL(2,p) / C(SL(2,p))$

since $|SL(2,p)| = p(p^2-1)$

$|SL(2,2)| = 2 \times 3 = 6$ $|SL(2,3)| = 3 \times 8 = 24$

$|SL(2,4)| = 4 \times 15 = 60$ $|SL(2,5)| = 5 \times 24 = 120$ $|SL(2,7)| = 7 \times 48 = 336$

now $|PSL(2,p)| = \frac{|SL(2,p)|}{|C(SL(2,p))|}$

so $|PSL(2,2)| = 3$

$|PSL(2,3)| = 12$

$|PSL(2,5)| = 60$

$|PSL(2,7)| = 168$

•

e)

$$|GL(2,2)| = 3 \times 2 = 6$$

$$GL(2,2) = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix} \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix} \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$$

$$PSL(2,2) = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix} \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix} \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$$

is exactly the same as $GL(2,2)$ since $1 = -1$ in $\mathbb{Z}_2$

$$PSL(2,2)$$

The center of $SL(2,2)$ is a trivial Group $\{e\}$, so $PSL(2,2) \cong SL(2,2)$

$$\text{So } PSL(2,2) \cong S_3$$

$$\text{Matrix } \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix} \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} \begin{pmatrix} 1 & 0 \\ 1 & 1 \end{pmatrix} \begin{pmatrix} 0 & 1 \\ 1 & 1 \end{pmatrix} \begin{pmatrix} 1 & 1 \\ 1 & 0 \end{pmatrix}$$

$$\text{element is } S_3 \quad e \quad (12) \quad (23) \quad (13) \quad (123) \quad (132)$$

f)

$$|PSL(2,3)| = 12$$

it has the same number of Elements

$$\text{as } A_4 \text{ since } |A_4| = \frac{1}{2} |S_4| = \frac{1}{2} 24 = 12$$

consider the carrier space of $SL(2,3) (\mathbb{Z}_3)^2$

There are 8 nonzero elements $(1,0) (0,1) (1,1) (2,1) (1,2)$

$(2,0) (0,2) (2,2)$, there are 4 lines connected by

each of two points, like $(1,0) - (2,0)$, $(1,1) - (2,2)$ - -

and here is a homomorphism between $SL(2,3)$, S_4

$$SL(2,3)/\ker \phi = \text{Image}(\phi) \quad \text{Image}(\phi) \text{ must be a subgroup of } S_4, \text{ with order 8,}$$

$$PSL(2,3) \cong \text{Image}(\phi) \quad \text{the only one is } A_4$$

Problem 2.

$$Q_4 = \{\pm e, \pm I, \pm J, \pm K\}.$$

the classes are $\{e\} \{-e\} \{\pm I\} \{\pm J\} \{\pm K\}$

Theorem 1 states: $M_1\text{-irrep} = \frac{|G|}{|C(G)|} = 4$

Theorem 2 states: $M_{\text{irrep}} = \# \text{Classes} = 5$

Theorem 3 states: $\sum_i \chi_i^2 [\dim(\mathbb{R}^{(i)})]^2 = |G|$

so there are 4 1 dimensional irrep and 1 2 dimensional irrep.

	$\chi(e)$	$\chi(-e)$	$\chi(\pm I)$	$\chi(\pm J)$	$\chi(\pm K)$
$\mathbb{R}_1^{(1)}$	1	1	1	1	1
$\mathbb{R}_1^{(2)}$	1	1	-1	-1	1
$\mathbb{R}_1^{(3)}$	1	-1	1	-1	1
$\mathbb{R}_1^{(4)}$	1	-1	-1	1	1
$\mathbb{R}_2^{(1)}$	2	0	0	0	-2

the 2 dimensional irrep is essentially Pauli Matrices, with minor changes

$$\pm I = \pm i\sigma_1$$

$$\pm J = \pm i\sigma_2$$

$$\pm K = \pm i\sigma_3$$

the faithful irrep is defined as $M(g_1) = M(g_2) \Rightarrow g_1 = g_2$

only $\mathbb{R}_2^{(1)}$ the twodimensional irrep is faithful.

by FS Criterion we are able to find

all one-dimensional irrep is Real and

the two-dimensional irrep is PseudoReal.

Problem 3:

$$L(I) = \begin{pmatrix} 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 & 0 \\ 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \end{pmatrix}$$

$$L(K) = \begin{pmatrix} 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 & 0 \\ 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \end{pmatrix}$$

$$L(J) = \begin{pmatrix} 0 & 1 & 0 & 0 & 0 & 0 \\ 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 \end{pmatrix}$$

the multiplicity n_i of an irrep χ_i in a Representation

with character χ_j $n_i = \langle \chi_i, \chi_j \rangle = \frac{1}{|G|} \sum_{g \in G} \chi_i(g) \overline{\chi_j(g)}$

now substitute χ_i with the unit irrep $(1, 1, 1, 1, 1)$

$n_i = 1$, so its multiplicity is 1.